

Homework 5

Question 1. Please read chapter 8 of Chartrand et al. and write a couple sentences about a topic/example/concept that you found difficult or interesting and why?

I liked the part in 8.4 where it talks about where this meaning of the word combination comes from and Pascals letter. I like learning about the history of math.

Question 2. Each student at a certain university is given a 6-digit code (such as 123789 or 001122).

- (a) How many different codes are there? 10^6 since there are 6 choices of 10 digits per position
- (b) How many codes read the same forward and backward? 10^3 since both halves must match then that narrows our positions to half
- (c) How many codes contain only odd digits? 5^6 since 1,3,5,7,9 are the only odd $\mathbb{N}$ under 10
- (d) How many codes contain at least one even digit? $10^6 - 5^6$ since we already found all codes that are all odds than we can just get all other codes

Question 3. How many different 10-bit strings begin with 1011 or 0110? there are 2^6 combinations per beginning since the first 4 positions are already decided. For both it would be $2(2^6)$

Question 4. A password on a computer system consists of four characters, each of which is either a digit or a letter of the alphabet. Suppose that each password must contain at least one digit and at least one letter. How many different such passwords are there? $62^4 - 52^4 - 10^4$ Since without character constraints it is 62^4 but there cannot be a case where all are letters (-52^4) nor a case where all are digits (-10^4)

Question 5. A total of 70 students who go to football, basketball or hockey games on a regular basis are surveyed as to which of these three events they attend. They responded:

- 38 students go to football games.
- 38 students go to basketball games.
- 35 students go to hockey games.
- 17 students go to both football and basketball games.
- 15 students go to both football and hockey games.
- 16 students go to both basketball and hockey games.

How many go to all three?

7 since there are a total of 70 people but there are 111 games attended, 48 (17+15+16) of those games are the result of people going to two separate games or more. That means there are 41 (111-70) games being attended by someone that has attended 1 game before, but there are 48 games that are the result of repeat game attendences which gives us 7 (48-41) games that must have been attended as a third game.

Question 6. How many people must be present to guarantee that

- (a) at least two have the same birthday?
- (b) at least two of their birthdays are in the same month?
- (c) at least three of their birthdays are in one of the months January, February, March, April or at least four of their birthdays are in one of the remaining months?

Assuming it is not a leap year*

- (a) You would need 366 since either the last birthday would fall on another or if not, then there is at least one day with multiple birthdays
- (b) You would need 13 people to guarantee that at least 2 birthdays are in the same month since there a 12 months, a thirteenth person would have to be the second in a month or if not then there is already another month that already has 2 or more birthdays in it

- (c) You would need 33 people to guarantee that either condition would be met. If we took the maxes of each condition without being a guarantee we would get $8 + 24 = 32$. An extra person (33) would guarantee that either condition would or would already be met.

Question 7. How many different 8-bit strings contain exactly 5 1s?

56 since $\frac{8!}{5!(8-5)!} = 56$

Question 8. Show that $\binom{2n}{2} = 2\binom{n}{2} + n^2$.

$\binom{2n}{2}$ can be stated as "select a combination of 2 things from a 2n things". $2\binom{n}{2} + n^2$ can be stated as "select a combination of 2 things from n things twice then select 1 thing from n twice". Since that can be simplified to "select 2 from 2n" and to fulfill that, you would have already fulfilled "select 1 from n twice" so it can be simplified to $\binom{2n}{2}$ which is equal $\binom{2n}{2}$

Question 9. A student committee is to consist of 3 seniors and 4 juniors. A total of 6 seniors and 8 juniors have volunteered to serve on the committee. How many committees are possible?

There are $\binom{6}{3} \cdot \binom{8}{4}$ possible, since we just get the possible choices for group one and multiply it by the combinations for group 2

Question 10. A total of 5 seniors, 3 juniors and 4 sophomores have volunteered to serve on a 4-person committee. How many committees are possible if

- (a) there is no other restriction on membership for the committee?
- (b) at least one senior, one junior and one sophomore must serve on the committee?
- (c) at least 3 seniors must serve on the committee?
- (d) at least one senior must serve on the committee?

(a) $\binom{12}{4}$

(b) $\left(\binom{5}{2}\binom{3}{1}\binom{4}{1}\right) + \left(\binom{5}{1}\binom{3}{2}\binom{4}{1}\right) + \left(\binom{5}{1}\binom{3}{1}\binom{4}{2}\right)$ add all cases where one class has two and others have one together

(c) $\binom{5}{4} + \binom{5}{3}\binom{7}{1}$ cases where all are seniors + cases where one is not a senior

(d) $\binom{12}{4} - \binom{7}{4}$ total cases - cases where there are no seniors